

Wheat From Chaff: Meta-Analysis As Quantitative Literature Review

T.D. Stanley

In an era characterized by the expansion of research publications and an avalanche of information, balanced and critical literature reviews serve a crucial function. They act as intelligent agents searching through mountains of potentially contradictory research to uncover the nuggets of knowledge that lie buried underneath. However, not all reviews are balanced. It is common in literature reviews for the author to dismiss certain studies as “misspecified,” a judgement that too often seems to be made in a subjective and selective manner for the purpose of disqualifying findings which run contrary to the beliefs of the author. Although concerns about the distorting effects of misspecifications are quite justified, the subjective and selective reference to misspecifications as a way of discounting unwelcome empirical evidence is not.

The purpose of this essay is to discuss the potential of meta-analysis to summarize, evaluate and analyze empirical economic research. Meta-analysis is a body of statistical methods that have been found useful in reviewing and evaluating empirical research results. If a number of independent studies have been conducted on a particular subject, using different data sets and methods, then combining their results can furnish more insight and greater explanatory power than the mere listing of the individual results.

In particular, meta-regression analysis is a form of meta-analysis especially designed to investigate empirical research in economics (Stanley and Jarrell, 1989; Jarrell and Stanley, 1990). In a meta-regression analysis, the dependent variable is a summary statistic, perhaps a regression parameter, drawn from each study, while the independent variables may include characteristics of the method, design and

■ *T.D. Stanley is Professor of Economics and Business, Hendrix College, Conway, Arkansas. His e-mail address is <stanley@hendrix.edu>.*

data used in these studies. Thus, meta-regression analysis can identify the extent to which the particular choice of methods, design and data affect reported results. Meta-regression analysis can help to explain the wide study-to-study variation found among research findings and offer specific reasons, based on the studies themselves, why the evidence on a certain question may appear contradictory or overly varied. Such studies can also suggest potentially fruitful lines for future inquiry and offer a prediction of the results that such new research will find.

This paper begins with an overview of the widespread use of meta-analysis in fields other than economics, like the assessment of clinical medical trials, and within economics. I will then sketch the process of conducting a meta-regression analysis, focusing in particular on evaluating the empirical evidence on Ricardian equivalence, but using other examples as well. This approach offers a useful framework for discussing the strengths and weaknesses of meta-analysis. The most important strength of meta-analysis is that it moves literature reviews away from casual judgements about “good” studies that deserve attention and “poor” studies that should be set aside, and instead provides a replicable statistical framework for summarizing and interpreting the full range of evidence.

The Widespread Use of Meta-Analysis

Over the past two decades, there have been hundreds of meta-analyses in the medical and social sciences, often revealing the unexpected and providing order to areas infused with controversy. Indeed, meta-analysis has become the accepted practice for evaluating the current flood of conflicting scientific evidence. It is “how science takes stock” (Hunt, 1997).

Many prominent examples involve clinical trials of new drugs and medical treatments. Such trials are notoriously expensive, both in terms of dollars and lives. Yet individual trials are rarely definitive (Zivin, 2000). Meta-analysis has been found to be invaluable in making the most efficient use of this vital experimental data, thereby saving lives (Antman et al., 1992). For example, streptokinase has long been investigated as a potential treatment of heart attacks for its ability to dissolve blood clots. For two decades, randomized trials were undertaken without achieving any clear recommendation, until three meta-analyses were published. “Although the results of the individual clinical trials had been contradictory and unreliable, all three meta-analyses found that when the trials were combined, they showed that the clot-dissolving agents almost certainly reduced the risk of death by a considerable margin” (Hunt, 1997, p. 86).

The medical profession was reluctant to rely on meta-analyses rather than on conventional clinical trials alone, just as some economists are reluctant to trust a meta-analysis, which mixes what they might consider to be “good” and “bad” studies, as much as they trust the individual study that they prefer. But for the past decade, the American Statistical Association has supported and encouraged the use

of meta-analyses, even when they are based on small studies (Hunt, 1997, p. 96).¹ “Quantitative research synthesis—*meta-analysis*—has gained increasing use in the past 20 years, and rightly so. . . . Now that meta-analysis is available, a series of small studies of related questions, each with somewhat different treatment protocols and cohorts of subjects—when summarized with a quantitative synthesis—offers (at least the prospect of) an advantage over one large study with a single protocol of comparable cost. . . .” (Gaver, 1993). The value of meta-analysis is now widely recognized. “Over time, however, acceptance has grown, and it is one of many accepted procedures used in science. Now, it is hard to find anyone who objects to its proper use” (Harris Cooper, quoted in Hoffert, 1997, p. 1).

There are many examples where meta-analyses clarified a controversial area of research. For example, meta-analysis has been used to establish a connection between exposure to TV violence and aggressive behavior (Paik and Comstock, 1994), the efficacy of coronary bypass surgery (Held, Yusuf and Furberg, 1989), the risk of secondhand smoke (He et al., 1999), and the effectiveness of spending more money on schools (Hedges, Laine and Greenwald, 1994; Krueger, 1999). Many of the “new” scientific findings periodically reported by the media on various treatments, drugs and supplements are gleaned from the latest meta-analysis.

Meta-analysis has been successfully employed in hundreds of applications throughout the social and medical sciences. Some prominent examples of meta-analysis in economics appear in Table 1. High returns in the advancement of empirical understanding await economic researchers willing to develop and apply meta-analysis.

Meta-Regression of Econometric Evidence: Testing Ricardian Equivalence

How should a reviewer conduct a meta-analysis? What problems or considerations lie along the way? The following discussion offers a broad outline of the process involved in conducting a specific kind of meta-analysis—a meta-regression analysis.² To demonstrate the possibilities and to illustrate “the steps” taken in conducting a meta-analysis, I will focus on a meta-analysis of Ricardian equivalence, based upon Stanley (1998).

Ricardian equivalence makes the claim that the government budget balance will have no effect on the economy. The argument underlying this claim is that if

¹ Roughly a decade ago, in a position paper distributed to regional officers of the American Statistical Association, Thomas Louis’s first challenge for statistics concerned meta-analysis. “Developing and applying methods of combining evidence and of assessing the weak points in the combining process, pose major challenges and opportunities for the 1990s” (American Statistical Association, 1989).

² Readers interested in greater technical details should consult other works: for example, Hedges and Olkin (1985), Hunter and Schmidt (1990), Stanley and Jarrell (1989, 1998), Jarrell and Stanley (1990), and Stanley (1998, 2000a, b).

Table 1
Examples of Meta-analysis in Economics

<i>Study</i>	<i>Subject</i>
Stanley and Jarrell (1989)	introduction to meta-regression analysis
Jarrell and Stanley (1990)	union wage premiums
Smith and Kaoru (1990)	recreation benefits
Weitzman and Kruse (1990)	profit sharing and productivity
Phillips (1994)	education and productivity
Button (1995)	transportation benefits
Card and Krueger (1995)	minimum wage effects
Phillips and Goss (1995)	taxes and local development
Button and Kerr (1996)	urban traffic congestion
Espey (1996; 1998)	gasoline demand elasticities
Loomis and White (1996)	benefits of endangered species
Doucoulagos (1997)	demand for Australian labor
Baaijens, Nijkamp and Montfort (1998)	regional multipliers
Stanley and Jarrell (1998)	gender wage gap
Stanley (1998)	tests of Ricardian equivalence
Ashenfelter et al. (1999)	returns to education
Stanley (2000b)	tests of the Lucas critique

people correctly anticipate future tax consequences of deficit spending and save accordingly, the government's dissaving will be exactly offset. David Ricardo did not believe that people actually behaved according to Ricardian equivalence (Ricardo, 1820; O'Driscoll, 1977), but then, why should his beliefs be allowed to spoil a good theory?

The predictions of Ricardian equivalence are empirically testable, and many such tests can be found in the economic literature. However, researchers who have reviewed this evidence in a conventional but unsystematic manner have concluded that "results are all over the map" (Barro, 1989, p. 49) or that "the empirical evidence is extremely ambiguous" (Perasso, 1987, p. 672). In contrast, the meta-analysis discussed below draws a rather clear picture from these tests of Ricardian equivalence. Thus, it illustrates the usefulness of this approach to the field.

Step #1: Include All Relevant Studies from a Standard Database

Meta-analysis begins with a resolute emphasis on including all studies, published or not, as a way of reducing potential biases introduced by any nonrandom selection of studies. A computer search of standard databases—for example, EconLit and Dissertation Abstracts—is clearly the place to start. Best meta-analytic practice gives the details of such a computer search, including the years covered, to allow easy replication by others (Beaman, 1991). Some meta-analysts make the additional effort of writing to every academic department and research institute for working papers and the relevant contents of their "file drawers."

Notwithstanding this broad presumption of inclusiveness, defining the appro-

priate population of studies can be tricky. Should different effects be considered together or separately? For example, one part of the literature on Ricardian equivalence considers the extent to which budget deficits affect consumption, while another part considers whether budget deficits affect interest rates. What about studies using data from different countries? Or studies employing different econometric methods?

The answer to such questions depends upon the issue being studied. However, when in doubt, it is best to err on the side of inclusion. After all, one function of the meta-regression analysis is to obtain estimates of how such research choices influence the results. Differences in quality, data or methods do not provide a valid justification for omitting studies. Rather, such difference provide the underlying rationale for doing a meta-regression analysis in the first place.

My meta-analysis of Ricardian equivalence began by searching EconLit's CD-ROM 1980–95 for any reference to “Ricardian equivalence” in all fields. A review of the 110 references to Ricardian equivalence netted 28 empirical studies. All 28 empirical studies of Ricardian equivalence were included in the meta-analysis.

Step #2: Choose a Summary Statistic and Reduce the Evidence to a Common Metric

After reducing the sample of studies to those that contain some relevant empirical estimate, test or finding, the next step is to identify important characteristics of the studies and to code them. This step can be difficult and time-consuming. Indeed, coding can represent as much as 90–95 percent of the work involved in conducting a meta-analysis (Hunter and Schmidt, 1990).

The best place to start is by deciding on what summary measure the researcher wishes to meta-analyze. For some fields, the choice of summary measures is clear, in others, some transformation of reported statistics are required. Regression coefficients, elasticities, wage gaps, t-values and the results of other statistical tests are all suitable choices.³

To summarize the evidence from different statistical tests, it is necessary to transform the summary statistics to a common and comparable metric. Gene Glass (1976, 1977), who launched the modern study of meta-analysis and popularized the name “meta-analysis,” introduced the concept of “effect size” to integrate the findings of an experimental literature. The original idea of effect size applied to experimental studies that involved a treatment group and a control group. In this context, effect size is defined as the average outcome of the treatment group minus the average outcome of the control group, divided by the standard deviation of the

³ For examples of the use of different statistics, see Stanley and Jarrell (1989), Jarrell and Stanley (1990), Card and Krueger (1995), Espey (1996), Stanley and Jarrell (1998), Stanley (1998, 2000a). There are also a few technical considerations about what summary variables to choose (Stanley and Jarrell, 1989). Such considerations include the type of distribution that the chosen summary statistic is likely to have, its potential heteroskedasticity, and its relation to a study's degrees of freedom.

control group. Effect size allows diverse studies, potentially even with different outcome variables, to be compared directly on the same dimensionless scale. Researchers in education and psychology often use effect size as an alternative to statistical significance, interpreting effect sizes of .2 as “small,” .5 “medium” and .8 “large” (Cohen, 1988).⁴

Beyond the convenience that such a summary statistic provides, its great advantage lies in the increase in power that follows from its aggregation. When the effect sizes from K independent studies are combined together, say in a simple average, statistical power increases corresponding to the sum of the sample sizes across all K studies. It is this increase in statistical power that motivates much of the development and application of meta-analysis. Uncovering statistical significance in a controversial area brimming with diverse and ambiguous findings has been the objective of many meta-analyses in the social sciences and medical research.

Statisticians have since improved upon Glass’s notion of effect size and have greatly widened its range of applicability (Hedges and Olkin, 1985). Nonetheless, Glass’s idea of using a common denominator remains central to meta-analysis.

In the case of testing Ricardian equivalence, individual studies use an econometric model to explain consumption expenditures, interest rates or the exchange rate. Researchers then estimate these econometric models both with and without the multiple parameter restrictions implied by Ricardian equivalence (Stanley, 1998, p. 715; Leiderman and Blejer, 1988). The particular statistical tests used (likelihood ratio, Lagrange multiplier, or Wald) vary from study to study and are distributed either as chi-square or F . However, both chi-square and F -distributions depend on the degrees of freedom available, which differ across studies, and therefore cannot be directly compared, summarized or analyzed. Thus, some transformation to a common metric is needed. Conversion to the standard normal is the most obvious and practical way to deal with these incompatible distributions.⁵ Once converted to a standard normal variable, results across the literature may be compared on a common scale and analyzed meaningfully. Large positive values of the test statistic Z_i reflect stronger statistical evidence against a study’s null hypothesis, while values near zero, or negative, are more supportive.

Normalizing the 28 tests of Ricardian equivalence reveals a large average effect, with an average test statistic (after converting all individual studies to a standard normal variable) of $Z = 2.41$, which is statistically significant at the .01

⁴ Though Cohen’s (1988) characterization of effect sizes is arbitrary, it provides a useful guideline, similar to the conventional and arbitrary choice of the .05 level of significance. Of course, if the researcher knows the error loss function, or equivalently the costs-benefits, practical significance may be defined accordingly.

⁵ Lipák’s (1958) inverse normal method finds the associated standard normal value, Z_p , by equating the cumulative probability of each calculated F or χ^2 to the cumulative standard normal probability for Z_i . Also, the Handbook of Mathematical Functions (Abramowitz and Stegun, 1964) provides convenient approximation formulas that convert both F s and χ^2 s to the standard normal distribution (for discussion, see Stanley, 1998, p. 716).

level. In other words, the average statistical test in this literature lies roughly two and a half standard deviations apart from Ricardian equivalence. Because Ricardian equivalence postulates the absence of any significant effect from deficit financing, it is strongly rejected by the empirical literature as a whole.

However, there are potential problems with using any such simple descriptive statistic to interpret diverse, often ambiguous, empirical economic literatures. Are there misspecifications or publication biases that elude simple statistical summary? Excessive variation among the reported results casts doubt on the adequacy of any simple descriptive statistic in representing a diverse literature. Among tests of Ricardian equivalence, there is a great deal of variation in the estimates of the normalized test statistic Z . The range from the lowest estimate of Z to the highest is 8.63; the standard deviation of Z is 1.96. Indeed, strictly random experimental variation should produce normalized values with a unit variance; that is, $\sigma_Z^2 = 1$.⁶ Tests of Ricardian equivalence clearly reveal excess variation at the .01 level of significance. This larger variation begs explanation. Explaining excess research variation is the purpose of the following two steps.

Step #3: Choose Moderator Variables

The summary statistic, reduced to a common metric, will serve as the dependent variable in a meta-regression analysis. The independent variables—often called “moderator variables”—are those study characteristics that are thought to be consequential. There is some latitude for identifying what these key characteristics should be. At a minimum, the meta-analyst will wish to code dummy variables for the use of different data sets and econometric modelling choices. However, because the number of studies is limited and most studies entail a unique combination of techniques, independent variables, data, time periods and other research choices, not every study characteristic can be coded and analyzed. Nor should a researcher wish to do so. Variation due to minor modelling choices may be treated as part of the random study-to-study background.

By way of illustration, return to the example of the meta-analysis of Ricardian equivalence. A number of potential moderator variables, which primarily represent differences in econometric specification, were selected as shown in Table 2. These characteristics were chosen before statistical analyses were conducted.

The list includes the number of degrees of freedom in the study and the average year of the data. There are four variables for the model underlying the study; that is, whether the regression is based on a model of rational expectations; a permanent income model; a Keynesian model that uses disposable income; or a vector autoregressive model. One variable addresses whether the econometric model contains a lagged dependent variable; another captures whether the model

⁶ For outcome measures that have not been normalized, their averages should be weighted, leading to alternative tests for homogeneity (Greenland and Longnecker, 1992; Cooper and Hedges, 1994).

Table 2

Meta-Independent Variables for Ricardian Equivalence

df	= the degrees of freedom available to the Ricardian equivalence test.
Year	= the average year of the data that a study uses.
Expect	= 1 if a study is based primarily on 'rational' expectations.
Lag	= 1 if a study used a lagged dependent variable.
PIH	= 1 if study used a permanent income model.
Keynes	= 1 if a study used a Keynesian model with disposable income.
VAR	= 1 if a study used a vector autoregressive model.
Diff	= 1 if a model is expressed primarily in differences.
Spec	= the number of specification tests passed less those that failed.
Consume	= 1 if a study investigated consumption expenditures.
US	= 1 if a study used only U.S. data.

is primarily specified as differences.⁷ The number of specification tests passed was selected as a measure of a model's quality and proper specification. There were also dummy variables for whether the study used consumption as its dependent variable and whether it relied on U.S. data.⁸

A frequent problem occurs when more than one estimate (or test) of a desired effect is given in a study. For example, in two studies, Dalamagus (1992a, b) reports tests of Ricardian equivalence in each paper that cover 52 countries. Should these results be treated as 52 "studies" or as two "studies"? Since there are only 28 research papers, counting the two Dalamagus papers as equivalent to 104 separate "studies" would clearly give these two papers a disproportionate importance. To avoid giving undo weight to a single study, one estimate should be chosen or averaged from many comparable estimates. Otherwise, one researcher (or even a couple of papers) might dominate an entire field through sheer mass. Therefore, multiple-estimate studies are averaged. When possible, we choose the estimate that the author believes to be the best. But this strategy is not without flaws, since a researcher who publishes a comprehensive single paper on a subject can be underweighted compared to a researcher who publishes a number of closely related papers that each count as an individual study.⁹

⁷ I have long suspected that econometric models of time series need to include lags and/or differences to be correctly specified (Stanley, 1991, 1993, 2000a). Many published models contain substantial misspecification biases because they do not fully account for the usual nonstationarity of economic time series. Whether this claim is generally true or not, it is useful to include these as variables in the meta-analysis and let the actual research record decide.

⁸ Of course, other choices of moderator variables are possible. For example, there are many ways to describe variations in econometric model specifications. Moderator variables may be defined for individual authors who publish more than one study, for the school from where the author graduated or for the journal in which the research appears. Ultimately, the research record should be used to choose.

⁹ These issues too can be treated objectively and their effects estimated by meta-analysis. For example, Jarrell and Stanley (1990) use a separate dummy variable for each multiple-study author, thereby

Step #4: Conduct a Meta-regression Analysis

With some summary variable to serve as the dependent variable and the moderator variables to serve as the independent variables, the groundwork is now set to conduct a meta-regression analysis. The purpose of meta-regression analysis is to explain the study-to-study variation found in an empirical literature.

The results of the meta-regression analysis of Ricardian equivalence are shown in Table 3. Nine of the eleven variables listed in Table 2 have a significant effect on Z_i . Positive coefficients suggest that a certain study characteristic is associated with a more significant, non-Ricardian effect, while negative coefficients identify study characteristics more consistent with Ricardian equivalence. Whether a study uses differences or lags, either of which can help to deal with (likely) nonstationarity, has important effects on Ricardian equivalence test results. More direct indicators of proper specification affect results as well. The number of specification tests a given study passes and the degrees of freedom available are both significant and positively associated with the normalized test statistic. This positive association between the number of specification tests passed and the normalized test statistic implies that better-specified models are more likely to find non-Ricardian effects. The positive association between degrees of freedom and normalized test statistics is more subtly related to proper model specification but indicates that there is genuine non-Ricardian behavior in the economy. (This relationship is developed more fully in the following discussion of Step #5.) A number of additional study characteristics are also found to affect the evidence on Ricardian equivalence, as shown in Table 3. Together, the nine significant variables explain an impressive 83 percent of the variation among tests of Ricardian equivalence.

In the case of Ricardian equivalence, the empirical literature has focused on whether various parameter restrictions are valid. In other cases, the issue has been the size of a particular economic magnitude. As a further illustration of the potential of meta-regression analysis, let us turn to meta-analysis of gender wage differentials (Stanley and Jarrell, 1998). In a meta-analysis of 41 estimates, the average gender wage differential after adjusting for experience, education level, and whatever other variables the research saw as relevant was 33.6 percent, with a standard deviation of more than half that value. Our meta-regression analysis of the gender wage differential identifies and codes 29 factors, including four alternative measures of wages, three different data sets, various model specifications, 14 possible omitted variables, the unemployment rate, a time trend variable, and the gender of the researcher. Of these, twelve are significant, explaining 87 percent of the variation among estimated wage differentials.

Meta-regression analysis reveals that the manner in which wages are calculated is the most important factor affecting the estimated gender wage gap. When wages

isolating and estimating these authors' individual slants. Interestingly, a number of these multiple-study authors found union wage gaps significantly different from the literature as a whole.

Table 3

Meta-Regression Results for Ricardian Equivalence

<i>Count:</i>	<i>R:</i>	<i>R-squared:</i>	<i>Adj. R-squared:</i>	<i>Standard error:</i>
27	.91158	.83098	.7415	1.00013

Analysis of Variance Table

<i>Source</i>	<i>DF:</i>	<i>Sum Squares:</i>	<i>Mean Square:</i>	<i>F-test:</i>
REGRESSION	9	83.60319	9.28924	9.28674
RESIDUAL	17	17.00458	1.00027	$p = .0001$
TOTAL	26	100.60777		

Estimated Coefficients

<i>Variable:</i>	<i>Coefficient:</i>	<i>t-Value:</i>	<i>P-Value:</i>
INTERCEPT	3.70738		
Expect	-3.73467	5.43044	.0001
Diff	-1.99477	3.98366	.001
df	.00584	3.4171	.0033
Spec	.75792	3.32059	.004
VAR	2.00237	3.06024	.0071
Lag	-1.82133	3.04833	.0073
PIH	1.10014	2.34653	.0313
US	.98184	2.04499	.0567
Keynes	-1.26449	1.79727	.0901

are estimated from annual or weekly salaries, the estimates of the gender wage differential are much higher. Estimates of the gender wage differential are also much higher (by 28 percent) if researchers omit worker experience in their analysis. On the other hand, focusing on new labor market entrants or failing to correct for selection bias causes estimates of the gender wage differential to be much lower. Contrary to some concerns about the objectivity of researchers, male researchers tend to find a significantly larger gender wage differential than do their female colleagues. Perhaps researchers compensate for potential bias implicit in their own group membership by giving any latitude in design and interpretation to the opposite sex.

Lastly, this meta-regression analysis uncovers a significant time trend in estimates of the gender wage differential (Stanley and Jarrell, 1998). They are declining by nearly 1 percent per year. Although other researchers have observed a similar declining trend in raw, unadjusted gender gaps (Murphy and Welch, 1991; O'Neill and Polachek, 1993), meta-regression analysis filters out several socioeconomic trends affecting the gender wage differential as well as particular characteristics of different studies. As a result, the significant coefficient on the time trend

variable in a meta-regression analysis offers more powerful evidence of a reduction in the gender wage differential than do studies based on unadjusted gender gaps alone.

The large coefficients in the meta-regression analyses of the literature on Ricardian equivalence and the gender wage differential suggest that the results of studies in these areas are quite sensitive to choice of econometric models and approaches. Indeed, one can use the results of meta-regression analysis to forecast the results of a future study that has certain characteristics. In one case, a meta-regression analysis of the union wage premium by Jarrell and Stanley (1990) was coincidentally published in the same issue of the *Industrial and Labor Relations Review* as a new study of the union wage premium by Linneman, Wachter and Carter (1990). If the characteristics of the Linneman, Wachter and Carter study are plugged into the reported meta-regression equation, it accurately predicts the results of this study's estimate of the union wage premium to within 1.4 percent.

The results of meta-regression analysis can also discover patterns and relations that might otherwise remain shrouded in fog. If a certain variable is found to be important in the meta-regression analysis, then empirical studies in this area should include this variable. For example, a meta-analysis revealed significant interactions between the union wage premium and unemployment effects (Jarrell and Stanley, 1990). However, some previous studies of the union wage premium had not included an unemployment variable, and even Lewis's (1986) generally splendid narrative review of this literature does not control for the effects of unemployment.

Step #5: Subject the Meta-Regression Analysis to Specification Testing

Like any scientific statistical model, meta-regression analysis is fallible and a critic may reasonably question whether a meta-regression model is itself properly specified. This question can be addressed with specific statistical tests. For example, both the meta-regression analysis of Ricardian equivalence (Stanley, 1998) and of the gender wage difference (Stanley and Jarrell, 1998) pass standard specification tests for autocorrelation, heteroskedasticity, and generic misspecification. As a further test of specification, the meta-regression analysis of the gender gap in Stanley and Jarrell (1998) held out part of the sample and then tested the model's performance in out-of-sampling forecasting.

In general, there is reason to suspect that meta-regression analysis may be less problematic than conventional econometric analysis (Stanley and Jarrell, 1989). Economic research findings may show some pattern over time (Goldfarb, 1995); however, they are less likely to be nonstationary and their disturbances are less likely to be autocorrelated than the typical economic time series. Furthermore, independent statistical tests that have been normalized are more likely to satisfy the classical regression assumptions than raw economic data.

A meta-regression analysis of normalized test statistics lends itself to a further examination of its particular specification. Recall that the dependent variable has a standard normal distribution under the null hypothesis of no effect. In particular,

purely random, sampling error should have a variance equal to one. When all systematic variation due to misspecification and variation in econometric models and methods of the original studies is adequately represented by the meta-regression analysis model, the remaining error variance will be equal to one.¹⁰ In the meta-regression analysis of Ricardian equivalence, the null hypothesis that this error variance is equal to one cannot be rejected (Stanley, 1998, p. 722). As a result, it would be unreasonable to expect a better-fitting model of the controversial Ricardian equivalence literature.

By explicitly including the study's degrees of freedom (or equivalently, the sample size), meta-regression can offer an additional, independent test for the existence of an authentic empirical effect in the literature. If there is a genuine underlying empirical effect, rather than the exploitable artifacts of econometric misspecification, then the empirical literature should exhibit a significantly positive relation between the normalized test statistic which is used as the dependent variable and the study's degrees of freedom. The absence of a positive relationship between degrees of freedom and these normalized test statistics raises questions about the presence of any authentic empirical phenomenon in the literature.¹¹

This association of power with sample size is well-known (Cohen, 1988; Murphy and Myers, 1998). In fact, Poirier (1995, p. 367) uses this property as his definition of "test consistency." Tests of parameter restrictions (including likelihood ratio, Lagrange multiplier, and Wald tests) can also be shown to have greater power as the available degrees of freedom increase (Poirier, p. 502).

It is important to remember that this relation of power to degrees of freedom exists only if there is a genuine, underlying empirical effect. Biases due to misspecification in the original studies will exhibit no systematic relation to degrees of freedom; after all, there is no reason to suspect that specification errors will be larger (or smaller) in studies with large samples. Also, this specification test is independent of the outcome of specification testing in the original studies or the weight of evidence as revealed by the meta-analysis.

To understand the implications of this specification test, return to the meta-regression results on Ricardian equivalence reported in Table 3. Not only does the literature reveal large average non-Ricardian effects, but these results are systematically larger for those models using greater degrees of freedom even

¹⁰ This test assumes a fixed-effects model.

¹¹ The importance of the degrees of freedom derives from its relation to the power and from the asymmetry of classical statistical testing. When the null hypothesis of some statistical test is true, statistically significant test results will be observed randomly and infrequently, 5 percent of the time. Because classical (Neyman-Pearson) hypothesis testing calculates the rejection rule for a fixed alpha conditional on the sample size (or degrees of freedom), a true null hypothesis is no more, nor less, likely to be rejected as the sample size increases. However, this is not the case should the null be false. If H_0 is false, then the probability of correctly rejecting H_0 (that is, statistical power) is positively associated with the sample size or, more precisely, its square root.

after adjusting for other study characteristics.¹² This result should make it considerably more difficult for defenders of Ricardian equivalence to use claims of misspecification bias as a convenient excuse to rationalize away unflattering empirical evidence.

In a study of the empirical literature on the impacts of the minimum wage on employment, Card and Krueger (1995) also exploit this relationship between the size of the test statistic and its degrees of freedom. In this minimum wage literature, they find a negative, but insignificant, relation between a study's degrees of freedom and its t-statistic. They interpret the absence of a positive relation largely as an indicator of publication bias (p. 242): "[S]tudies in the literature have been affected by specification-searching and publication biases, induced by editors' and authors' tendencies to look for negative and statistically significant estimates of the employment effect of minimum wages." I prefer to view such an absence as a reason to question the authenticity of the underlying empirical effect, without differentiating between potential specification and publication biases. But then, Card and Krueger (p. 242) also explicitly recognize the broader interpretations. "Structural change is one possibility: the true effect of minimum wage may have declined over time. . . ." To differentiate among potential interpretations—nonrandom specification biases, publication biases or fortuitous structural change—of this lack of positive association between t-values and degrees of freedom, a more complex meta-regression model could be used to test for structural change. But unless or until such a study is completed, the negative association between degrees of freedom and t-values in this literature is a warning not to place much trust in any summary statistics of minimum wage effects.

Meta-Analysis vs. Narrative Literature Reviews

Conventional narrative literature reviews containing no explicit meta-analysis can represent some of the best research that economics has to offer. For example, H. Gregg Lewis's (1986) review of union wage effects is masterful, comprehensive, fair and enlightening. Likewise, Thomas Mayer (1993) provides an excellent and even-handed review of the Lucas critique. Yet even the best narrative reviews can be improved through meta-analysis (Jarrell and Stanley, 1990; Stanley, 2000a).¹³ Although meta-analysis can involve questionable decisions or controversial findings, these problems are generally much worse in narrative literature reviews.

¹² Because a variety of different statistical tests were first converted to the normal, there is no precise expectation for the coefficient on degrees of freedom—beyond being positive. If, on the other hand, all tests were t's, then coefficient on the logarithm of the square root of the sample size should be exactly one when regressed on of the logarithm of t (Card and Krueger, 1995, p. 240).

¹³ The discussion of narrative literature reviews throughout this section expands upon Stanley (1998).

Problems of Narrative Literature Reviews

A first problem that is common to narrative literature reviews is that the reviewer can introduce a substantial bias by omitting portions of the literature, usually on alleged methodological grounds—"methodological selection bias." When reviewing an empirical literature, many economists seem to become Bayesian, holding strong priors formed on the basis of theoretical considerations and raising high barriers to any contrary empirical evidence.¹⁴ "Believing is seeing" (Demsetz, 1974, p. 164).

These problems arise in many narrative literature reviews. Because this paper has focused on the empirical literature on Ricardian equivalence, it may be useful to illustrate the problems of narrative literature reviews with two leading examples in this area: Seater's (1993) very thorough review of the literature on Ricardian equivalence in the *Journal of Economic Literature* and Barro's (1989) essay on the evidence for Ricardian equivalence which appeared in this journal.

Both of these reviews judge that a number of empirical studies are untrustworthy for reasons of specification and methodology, and then find that the remaining trustworthy studies offer support for Ricardian equivalence. For example, Seater (1993, p. 143) writes: "Although there is much evidence appearing to reject Ricardian equivalence, . . . [w]hen attention is restricted to the methodologically sound studies, it is difficult to find statistically significant effects of debt, suggesting that Ricardian equivalence holds approximately." Barro (1989) uses potential problems of simultaneity and the endogeneity of deficits as a reason to replace econometric tests, which look at the impact of government deficits on consumption and savings, with studies that look at the impact on interest rates and with case study evidence.¹⁵ However, these potential specification problems apply equally to the supportive evidence from interest rate studies that Barro embraces. Ironically, the problems of simultaneity and endogeneity bias, used by Barro to dismiss consumption studies, are most likely to work in favor of Ricardian equivalence, not against it. As Bernheim (1987, p. 282) argues, "Shocks to consumption may be correlated with shocks to income, which in turn raise tax revenue (lower deficits). Thus, there is a natural bias in favor of Ricardian equivalence, even in a Keynesian world."

Of course, specification and methodological issues can be raised about any empirical study. It is not enough to say that some studies have potential problems; all studies have potential problems. The issue is to sort out and estimate, when

¹⁴ It is encouraging to observe that in the meta-regression analysis of the gender wage differential, male researchers found significantly larger discrimination against women than female researchers. However, researcher gender may be a poor proxy for a priori theoretical beliefs.

¹⁵ Using anecdotal evidence, or "natural experiments," only compounds selection biases. By selectively choosing which "natural experiment" to report, an author can be guaranteed the appearance of empirical confirmation.

possible, the impact of these methodological problems across the full range of studies.

With the benefit of the earlier meta-regression analysis on the Ricardian equivalence literature, some of Seater's and Barro's concerns seem misplaced. For example, Seater (1993) discounts studies that employ a Keynesian framework. Ironically, the earlier meta-regression analysis reported in Table 3 finds that these studies are marginally less likely to reject Ricardian equivalence, perhaps reflecting Bernheim's (1987) observations. Similarly, Barro's methodological concerns about the biases of consumption studies are not evidenced in the empirical record. Meta-regression results show that studies which focus on consumption are not significantly different than other studies (Stanley, 1998). In addition, both Seater and Barro are far less likely to raise methodological issues about the studies that tend to support Ricardian equivalence than about the studies that find contrary evidence—even though very similar methodological issues exist.

These issues of choosing between trustworthy and untrustworthy studies may be especially important in the Ricardian equivalence literature, since in that literature, a finding that government budget balance or deficits have no statistically significant effect on consumption or interest rates is taken as confirmation of the Ricardian equivalence hypothesis. The combination of low statistical power in practical samples and random specification biases can easily cause a majority of tests in *any* area of empirical research to find no effect where there are, in fact, important effects. The meta-regression analysis of this area suggests that it is the better-specified models (that is, those that pass additional specification testing) and the more accurate tests (that is, those with larger sample sizes) that are more likely, not less likely, to reject Ricardian equivalence. Rather than relying on casual methodological speculation to sift through a body of studies, meta-regression analysis allows the empirical estimation and testing of such speculation.

The Vote-Counting Approach

Some conventional narrative reviews do accumulate all of the available studies and put them carefully into a table, often with results categorized as positive, negative, or uncertain. This is sometimes referred to as “vote-counting,” where each study gets a vote. Among meta-analysts, vote-counting is considered to be biased and obsolete (Hedges and Oklin, 1985; Hunt, 1997). After all, when statistical tests have low power, a common occurrence in econometric applications, vote-counting schemes are biased in favor of finding no significant effect. In the case of Ricardian equivalence, for example, vote-counting is biased in favor of Ricardian equivalence, because the absence of effects is seen to support Ricardian equivalence. The likelihood that vote-counting will come to the wrong decision can be quite large, and, counterintuitively, it may even increase with added research (Hedges and Oklin, pp. 50–51).

DeHaan and Zelhorst (1988) provide a clear example of the problems inherent in vote-counting in the Ricardian equivalence literature. They present a table

listing 49 separate tests of this literature. Thirty tests (or 61 percent) are classified as “falsifying” Ricardian equivalence. Nonetheless, the authors infer (p. 409): “As follows from the table, the empirical evidence provides mixed *support* for debt neutrality theory” (emphasis added). One wonders how low the percentage would have to fall before it would no longer qualify as “mixed support,” but rather as “opposition.” Alternatively, one might argue that when empirical tests are weak and misspecification frequent, it would not be surprising if 39 percent of the studies fail to find an effect that is actually there. Even if the *majority* of the reported tests failed to find any empirical effect, it might say more about the prevalence of misspecification than about actual “positive” support for Ricardian equivalence. A simple vote-counting of studies is often misleading.

Limitations of Meta-Regression Analysis

Like any statistical tool, meta-analysis can be misused, and it has its own weaknesses. A number of these weaknesses have been mentioned or hinted at in earlier parts of this paper, but it might be useful to discuss these issues as a group here.

First, there can be disagreement over which study characteristics—or moderator variables—are the important ones to include in the analysis.

Second, when meta-regression analysis weights each published study equally, it risks overweighting the results of those who publish many smaller articles, each with a single result, compared to larger articles with a substantial number of results.

Third, a problem can arise in the interpretation of meta-regression analysis if all studies contain the same misspecification. In such a case, there is no way to distinguish and to estimate this common misspecification bias.

Fourth, there may be a problem of publication bias, where journal editors are more likely to publish some results—perhaps those that find statistically significant effects—than other results. Among meta-analysts this is called “file drawer” problem, because researchers may be more likely to consign insignificant results to the “file drawer” never to be published. If studies that fail to uncover statistically significant effects are less likely to be submitted to journals, or accepted for publication, then published results will tend to overstate the size and significance of experimental effects.¹⁶

Fifth, meta-analysis is sometimes criticized for considering all empirical studies, regardless of their quality.

In general, proponents of meta-analysis offer several reactions to this list of concerns. First and foremost, versions of these problems exist for any literature

¹⁶ Card and Krueger’s (1995) meta-analysis of the effects of changes in the minimum wage on employment is an insightful example of file drawer problem in economics. They observe a surprising number of studies that report negative employment effects with t-values quite close to the critical value of two. The clear implication is that published studies are not a random draw of estimated minimum wage effects.

review, not just for meta-analysis. For example, narrative literature reviews must also decide what study characteristics are relevant for discussion in the review. Narrative literature reviews also face a problem of how to weight individual studies or individual results from studies. If there is a misspecification across the literature, it will also be hard for a narrative review to spot and impossible to correct. Narrative literature reviews of published evidence are also subject to the problem of publication or file-drawer bias, while vote-counting literature reviews also face the problem of counting all studies, regardless of quality.

Proponents of meta-analysis then go on to make a stronger claim. A literature review that includes meta-analysis is likely to do a more satisfactory job in addressing all of these problems than is a narrative literature review.

In a meta-analysis, for example, discussions of which study characteristics are important can be examined statistically and choices can be compared. Indeed, estimating the effects of such differences is precisely the purpose of meta-regression analysis and the reason for its use of moderator variables. If there is a question about whether to weight studies individually or to weight individual results, a meta-regression analysis can try it both ways and see what actual difference it makes to our understanding of this research. To determine if the empirical literature is contaminated from nonrandom specification or publication bias, the meta-analyst can test whether the coefficient on the degrees of freedom variable is statistically positive. Such strategies are not available to the narrative reviewer. If certain studies are to be ruled out as being of low quality, meta-analysis impels the researcher to enunciate and code the specific characteristics that identifies the inferior nature of these studies. Studies should be omitted only by objective criteria that are applied evenly across the entire literature.

Ultimately, the advantage of meta-analysis is that all of these choices are made explicitly and their implications pursued statistically, allowing open discussion and testing. Even if meta-analysis is misused, inappropriate applications can be discussed and criticized on specific and technical grounds without descending to an argument about the motives of the author. Literature reviews will probably always maintain a strong narrative element. But a more widespread use of meta-regression analysis would make literature reviews both more convincing and more useful.

Conclusion

Economics is driven by theory. Discussions of theory dominate academic journals, shape empirical economic inquiry, and determine policy positions. Although the importance of theory is philosophically justifiable, economic science cannot progress without independent empirical testing. If economics is to learn from experience, objective interpretations of the empirical record must be encouraged and nurtured.

Meta-analysis provides a more formal and objective process of reviewing an

empirical literature. It employs conventional statistical methods and criteria to summarize and evaluate empirical economics. Unlike narrative reviews, meta-analysis uses objective procedures for the selection and analysis of studies. It avoids the inherent biases of vote-counting and methodological selection while offering additional, independent tests of significance and specification. Where conventional narrative reviews express ambiguity but mixed support for Ricardian equivalence, meta-analysis uncovers and corroborates a clear rejection.

The same arguments that justify a role for econometric techniques over an exclusively narrative discussion of economic history and anecdotes supports the use of statistical methods in the review of empirical literature—that is, meta-analysis. Both econometrics and meta-analysis have limitations and can be abused, but they also can be criticized objectively, refined and improved. Like econometrics, the application of meta-regression analysis is likely to reveal patterns and phenomena unimagined in previous narrative discourse.

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